

Data Structures & Algorithms - 30 Practice Questions

- Q1.** Define time and space complexity. Differentiate Big O, Theta and Omega notations.
- Q2.** Explain arrays and their advantages and disadvantages.
- Q3.** Write algorithms for insertion and deletion in arrays.
- Q4.** Differentiate arrays and linked lists.
- Q5.** Explain singly, doubly and circular linked lists.
- Q6.** Write an algorithm to reverse a linked list.
- Q7.** Explain stack operations and applications.
- Q8.** Implement stack using arrays and linked lists.
- Q9.** Explain queue, circular queue and deque.
- Q10.** Differentiate stack and queue with examples.
- Q11.** Explain recursion with an example.
- Q12.** Write recursive algorithm for factorial and Fibonacci.
- Q13.** Explain binary search and compare with linear search.
- Q14.** Explain bubble, selection and insertion sort.
- Q15.** Explain merge sort and its complexity.
- Q16.** Explain quick sort and worst-case scenario.
- Q17.** Compare merge sort and quick sort.
- Q18.** Explain binary trees and traversals.
- Q19.** Write preorder, inorder and postorder traversals.
- Q20.** Explain binary search tree operations.
- Q21.** What is AVL tree? Explain rotations.
- Q22.** Explain heap and heap sort.
- Q23.** Explain graph representations.
- Q24.** Differentiate BFS and DFS.
- Q25.** Explain Dijkstra's algorithm.
- Q26.** Explain minimum spanning tree and Prim's algorithm.
- Q27.** Explain Kruskal's algorithm.
- Q28.** What is hashing? Explain collision handling techniques.
- Q29.** Explain dynamic programming with an example.
- Q30.** Compare greedy algorithms and dynamic programming.